

Jens Laufer

Forward Deployed Engineer — autonomous AI agents and the harness that makes them reliable

Harness & Agentic Engineering · Forward Deployed · Python · LLM agents / MCP · RAG · Enterprise integration · Java / Spring Boot · Vue 3 · Docker

~29 years of software engineering (freelance since 1997), a second track in data science and machine learning — and since late 2025 harness and agentic engineering: the scaffolding around an LLM that turns a language model into a system you can leave running. Complex projects delivered agile and test-driven, always close to the client.

AVAILABLE immediately	WORKS remote only	REMOTE SHARE 100 % 0 % on-site
HOURLY RATE €100/h on-site €95/h remote · excl. VAT	BASED IN Karlstein am Main Germany	

FOCUS

~29 years of software engineering (since 1997), freelance throughout.

Java/Spring Boot microservices with Vue.js and React front-ends — end-to-end with Docker, Kubernetes and GitLab CI.

Strictly test-driven (TDD, JUnit, Mockito, Playwright/Selenium).

Second track in data science and machine learning — Udacity ML Engineer & Data Analyst, Kaggle, writer at Towards Data Science.

Broad industry exposure: banking/finance, insurance, automotive, telecommunications, travel, industry, retail, marketing and startups.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management
- Data analysis
- Prototyping
- Machine learning
- Jenkins plugin development
- Maven plugin development

SKILLS

LANGUAGES

- Java
- JavaScript
- TypeScript
- Python
- R
- Groovy

MICROSERVICES / BACKEND

- Spring Boot
- Spring
- Spring Data
- Spring Integration
- Spring Security
- OAuth2
- Microservices
- Feign
- Sleuth
- Zipkin
- Python Eve
- Python Flask
- Java EE
- EJB

FRONTEND

Vue.js

React

AngularJS

GWT

Quasar

HTML

CSS

Bootstrap

DEVOPS / CLOUD

Docker

Kubernetes

Rancher

GitLab CI

Jenkins

Maven

Grunt

npm

Artifactory

AWS (EC2, S3, SageMaker, VPC)

Microsoft Azure

DigitalOcean

TESTING

JUnit

TDD

Mockito

Selenium

DBUnit

Playwright

MACHINE LEARNING

scikit-learn

Keras

DeepLearning4J

CNN

Transfer Learning

Supervised/Unsupervised

k-means

PCA

t-SNE

Reinforcement Learning

OpenAI Gym

Topic Modeling

TOOLING / OTHER

Git

GitLab

Subversion

Jira

Confluence

Kibana

ANTLR

JavaCC

ZXing (QR)

PayPal API

RDF/XML/JSON

SAP (BAPI)

Phonegap

JBoss

Tomcat

PERSISTENCE / DATABASES

MySQL

MongoDB

PostgreSQL

Oracle

Neo4j

JPA

Hibernate

SECURITY / IAM

Keycloak

Spring Security

OAuth2

API / INTEGRATION

REST

Swagger

OpenAPI

SOAP

Webservices

CXF

XFire

Kafka

ServiceMix

DATA SCIENCE

pandas

numpy

dplyr

Jupyter

RMarkdown

ggplot

seaborn

Vega-Lite

A/B Testing

Webscraping

PROJECT HISTORY (SELECTION FROM 2009)

since 04/2026
ongoing

Fabrik — agent harness for parallel software delivery

Fabrik (fabrikhq.com) · Solytics GmbH · Remote — Software / AI
Harness engineering · Architecture · Full-stack engineering

A harness in which worker agents turn tickets into pull requests in parallel (fabrikhq.com). A **control plane** (FastAPI, Vue) spawns one agent container per repository through the Docker socket. Each agent runs a fixed loop — ticket sync (GitHub, GitLab, Jira), an actionability review that rejects vague tickets, a plan, test-first implementation, a **quality gate** in a clean sidecar container, then the PR; it never merges on its own. Energy budget per agent, feedback from PR comments back into the loop, one VPS per customer behind Caddy.

Claude / Anthropic API

Multi-Agent-Orchestration

Python

FastAPI

Vue 3

Docker

Caddy

GitHub Actions

SQLite

TDD

since 03/2026

ongoing

Autonomous coding pipeline — agents work a task queue unattended

agent-tasks · Solytics GmbH · Remote — Software / AI

Harness engineering · Agent engineering · DevOps

Queue-driven harness in which coding agents work through tasks with nobody at the keyboard. A task file names repository, scope, acceptance criteria and model; a runner on own hardware executes it in an isolated worktree, runs the **quality gate** (unit tests, build, review) before any push and opens a pull request — merging stays with a human. Model chosen per task against a 5-hour token budget, hard timeouts per run, at most one new pull request per repository per night. **759 tasks** written to date across the own product repositories.

Claude / Anthropic API

Python

Docker

TDD

systemd

since 10/2025

ongoing

Autonomous AI business partner — Harness & agentic engineering

Solytics GmbH · Remote — AI / Automation

Harness engineering · Agent engineering · AI system architecture · DevOps

A self-directing agent built on Claude that plans, writes code, tests, deploys and reports around the clock. Built as a **harness engineer** — the engineering around the model: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **observability** over every run, **budget and rate-limit control** with model routing, plus recovery when a run is killed mid-flight. More than 40 purpose-built checks catch the agent's mistakes before they reach a human. Event-driven over Telegram, GitHub and systemd; runs on own hardware behind a local multi-model proxy.

Claude / Anthropic API

LLM-Agenten

Agent Harness

Tool Use / MCP

RAG

Prompt Engineering

Python

systemd

Telegram Bot API

Docker

TDD

since 03/2026

ongoing

Verification and drift controls for LLM agents

Solytics GmbH · Remote — AI / Automation

Evaluation & verification · Reliability engineering

More than 40 purpose-built checks that test an agent's work against the world instead of against its own report: every link in an outgoing message is resolved before it is sent, promises the agent made are tracked until delivered, messages without an answer are flagged, pull-request mergeability is computed with git instead of read from a cached API value, and a test run that never started fails instead of passing silently. Each check fails closed — unknown is never reported as healthy — and has its own tests. Watchdogs over the agent's services and credentials, with a second alert path for when the primary channel is the one that broke.

Claude / Anthropic API

Python

Docker

TDD

since 03/2026

ongoing

LLM agent in the finance back office of a GmbH

Solytics GmbH · Remote — Finance / Compliance

Agent integration · Domain design · Back-end engineering

An agent that runs the bookkeeping of a German GmbH end to end, with a human approving every filing. It files bank statements and receipts, books double-entry on SKR04 through a purpose-built CLI, reconciles every account against the statement, and checks VAT and payroll returns against the journal before they are submitted — once catching a field that would have lost the input tax. Tools beneath it: an API for **XRechnung 3.0** and **ZUGFeRD 2.3** with five validation layers (Pydantic, BR-DE rules, arithmetic, XSD, Schematron), balance sheet, P&L, E-Bilanz and **DATEV export**.

Python

Pydantic

XRechnung / ZUGFeRD (EN 16931)

Polars

FastAPI

Docker

TDD

since 03/2026
ongoing

Launch Kit — multi-tenant identity and email platform

Launch Kit · Solytics GmbH · Remote — SaaS / IAM
Architecture · Back-end engineering · Security

OIDC-compliant auth server with email marketing as one Go binary and a Vue admin UI — a drop-in for Keycloak or Auth0. **OAuth2 with PKCE**, RS256 JWTs with a per-tenant RSA key, Login with Google, role-based access. Contacts, segments and campaigns with async sending, unsubscribe handling and GDPR consent. Strict tenant isolation; apps validate tokens over JWKS only and swap the provider by changing an environment variable. Runs the customer funnel of several own products.

- Go
- OpenID Connect
- OAuth2
- JWT
- Vue 3
- SQLite
- Docker
- Caddy
- Playwright

since 01/2026
ongoing

Paid browser extensions — FinGrab, PlaylistGrab, InvoiceGrab

FinGrab · Solytics GmbH · Remote — FinTech / Browser software
Full-stack engineering · Release & operations

A portfolio of browser extensions for Chrome (MV3) and Firefox from one codebase with WXT and TypeScript; FinGrab exports market data from Yahoo Finance as CSV and has around 600 users. **Live payments** through ExtPay and Stripe, subscription and one-time purchase. End-to-end tests with Playwright against the built package, a regression diff against the shipped version, publishing over the API to the Chrome Web Store and addons.mozilla.org, monitoring of reviews, subscriptions and the payroll.

- TypeScript
- JavaScript
- Browser extensions (Chrome MV3 / Firefox)
- WXT
- Stripe / ExtPay
- Playwright
- TDD

07/2024–06/2025
12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet
Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

- Python
- Python Eve
- MongoDB
- TDD
- Vue.js
- Docker
- GitLab
- GitLab CI
- REST
- Playwright

07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry
Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

- Java
- Spring Boot
- MongoDB
- JUnit
- TDD
- React
- Docker
- GitLab
- GitLab CI
- Spring Security
- Keycloak
- Mockito
- OpenAPI
- Swagger
- REST
- Microservices
- Playwright

01/2022–06/2022

6 months

Payment back-end migration Java EE → Spring Boot

Condor · Remote — Marketing

Backend engineering

Migrated an EJB-based Java EE payment back-end to Spring Boot and shipped it as a Docker container.

Java

Java EE

JPA

Spring Boot

MySQL

Docker

Mockito

JUnit

TDD

04/2021–11/2021

8 months

Software and data analysis in marketing

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Prototyping · Solution design · Data analysis

Server-side services in Java and Spring Boot: web-scraping microservices plus microsites (Vue.js, Quasar) with a Spring Boot back-end for calculators, surveys and questionnaire funnels; data analysis in R. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Feign), persisted in PostgreSQL and MongoDB. Built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes.

Scrum

Spring Boot

Docker

Kubernetes

Microservices

REST

R

Python

Java

OpenAPI

Feign

JUnit

TDD

Mockito

Vue.js

Quasar

MongoDB

dplyr

GitLab CI

GitLab

PostgreSQL

Python Eve

Python Flask

DigitalOcean

Webscraping

A/B Testing

Topic Modeling

01/2021–03/2021

3 months

Marketing campaign tooling

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Solution design

Tools that generate marketing campaigns and publish them to Google Ads and Twitter Ads: Java/Spring Boot microservices with REST interfaces (OpenAPI, Feign), Vue.js/Quasar front-end, storage in AWS S3. Designed and built in a microservices architecture. Built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes.

Scrum

Spring Boot

Docker

Kubernetes

Microservices

REST

Java

OpenAPI

Feign

JUnit

TDD

Mockito

Vue.js

Quasar

AWS S3

GitLab CI

GitLab

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin; built and maintained the GitLab CI pipelines.

Java

Spring Boot

Docker

Kubernetes

Microservices

REST

Spring Data

JPA

JUnit

TDD

Mockito

Vue.js

MySQL

MongoDB

HTML

CSS

GitLab

GitLab CI

Rancher

Kibana

Feign

Keycloak

Sleuth

Zipkin

07/2019–12/2019

6 months

Training a stock-trading agent

JiGaoTrading · Remote — Finance

Machine learning · Data analysis

An agent that trades autonomously on Chinese stock exchanges (reinforcement learning).

Reinforcement Learning

Python

R

Docker

AWS S3

OpenAI Gym

GitLab

seaborn

AWS SageMaker

MongoDB

04/2019–08/2019

5 months

Data analysis and modelling at a consumer cooperative

Co-operative Group (Coop) · Manchester — Finance, Retail

Machine learning · Data analysis

Exploratory data analysis, data cleaning, aggregation; unsupervised learning (clustering, feature reduction).

R

Python

Jupyter

RMarkdown

ggplot

scikit-learn

pandas

numpy

dplyr

k-means

PCA

t-SNE

Docker

Git

11/2018–03/2019

5 months

Image homogeneity of product niches from product photos

Sellics · Remote — E-commerce

Machine learning · Data analysis

A model that quantifies the image homogeneity of Amazon product niches to spot under-served niches early. Trained a ResNet50 CNN with transfer learning (Keras) and deployed it through a build pipeline as a Java microservice with DeepLearning4J into the existing landscape; prepared a Kafka data pipeline for continuous retraining.

Python

Keras

ResNet50

Transfer Learning

CNN

DeepLearning4J

Java

Kafka

Microservices

Docker

06/2018–10/2018

5 months

Aesthetic classification of Airbnb photos

AirDNA · Remote — Travel

Machine learning · Data analysis

A microservice that scores the quality of Airbnb photos: a Python scraper built an image database (~15k images), labelling via Amazon Mechanical Turk, exploratory data analysis in R; trained a CNN with transfer learning (Keras, ResNet50) on an AWS GPU instance and deployed it as a Spring Boot microservice with DeepLearning4J.

Java

Python

Keras

ResNet50

CNN

Transfer Learning

DeepLearning4J

Spring Boot

R

AWS

GPU

Jenkins

Amazon Mechanical Turk

Webscraping

Microservices

08/2017–03/2019

20 months

Connected Van — driver's logbook

Volkswagen · Remote — Automotive

Backend engineering

Microservices for vehicle geolocation and a fuel logbook.

Spring Boot

Docker

Kubernetes

Microservices

REST

Java

Maven

Swagger

OpenAPI

Feign

Sleuth

Zipkin

MySQL

MongoDB

Jira

Confluence

JUnit

TDD

Mockito

Microsoft Azure

Kibana

Spring

Git

Kafka

04/2017–07/2017

4 months

Invoice management platform (SME) — ML

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Machine learning · Data analysis

Data analysis; prototype for predicting incoming payments (ML).

Python R ggplot scikit-learn pandas numpy seaborn RMarkdown Jupyter Git

07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot Docker Kubernetes Microservices REST Java Maven Spring Security OAuth2
Spring Integration MongoDB Jira Confluence JUnit TDD Mockito Selenium AngularJS Grunt
npm Spring Git

07/2015–06/2015

5 months

Invoice management platform (SME) — debt-collection integration

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Microservice integrating a debt-collection provider, including a single-page admin front-end.

Spring Boot Docker Kubernetes Microservices REST Java Maven Spring Security OAuth2
Spring Integration MongoDB Jira Confluence JUnit TDD Mockito Selenium AngularJS Grunt
npm Spring Git

01/2015–02/2015

2 months

Invoice management platform (SME) — OAuth2 service

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

OAuth2 microservice with an administration front-end.

Spring Boot Docker Microservices REST Java Maven Spring Security OAuth2 MongoDB
AngularJS Git

09/2014–12/2014

4 months

Invoice management platform (SME) — prototype

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering · Solution design · Prototyping · Build & configuration management

Evaluation of microservices and SPA approaches, prototype for an invoice management platform.

Java JavaScript HTML CSS Spring Bootstrap AngularJS Grunt npm Spring Boot Docker
Kubernetes Microservices REST Maven Spring Security MongoDB Jira Confluence Git

01/2014–08/2014

8 months

Vehicle configurator prototype

Audi · Remote — Automotive

Full-stack engineering · Solution design · Prototyping · Build & configuration management

SPA on AngularJS (front-end) with a Spring back-end and a RESTful API; comparison of persistence approaches (MySQL, MongoDB, Neo4j).

Java JavaScript HTML CSS Bootstrap Spring Spring Security Spring Data MongoDB Neo4j
MySQL Maven Tomcat JUnit TDD Mockito Selenium Jira Git

06/2012–12/2013

19 months

Business services for a booking and reservation system

Deutsches Reisebüro (DER) · Frankfurt a. Main — Travel

Backend engineering

Implementation of business services as part of the re-build of a booking and reservation system.

Java JUnit TDD Mockito Spring SOAP Webservices Jira Maven

05/2011–04/2012

12 months

QR-code based payment solution

Paycoo Ltd. · London — Finance, Startup

Full-stack engineering · Solution design · Build & configuration management

Money transfer between accounts by scanning a QR code; hybrid app (GWT) embedded in native mobile apps for broad platform coverage.

Java JUnit TDD Mockito GWT Phonegap Spring Spring Security JPA REST ZXing
PayPal API Maven AWS EC2 AWS VPC Jenkins Artifactory

10/2010–02/2011

5 months

Entry testing (EntryCheck) of application server environments

Deutsche Telekom · Darmstadt — Telecommunications

Jenkins plugin development · Maven plugin development

Test environment for the technical entry testing of runtime environments (build and deployment) for a large Java EE enterprise application.

Java Maven Jenkins Groovy Subversion (SVN)

04/2010–09/2010

6 months

Web services on an ESB (mediation and rating)

1&1 · Montabaur — Telecommunications

Backend engineering

Software design, web services, maintenance of the rating engine; web services that hand invoices over to **SAP**.

Java Spring Spring Security JPA Hibernate Maven JUnit TDD Mockito DBUnit ServiceMix
Webservices CXF XFire SAP-Anbindung SAP BAPI JBoss Oracle Subversion (SVN)

05/2009–03/2010

11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance

Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.

Java Java EE IBM MDM (WCC) EJB JPA RDF XML Oracle Subversion (SVN)

CONTACT & CREDENTIALS

CONTACT & PROFILES

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LINKEDIN	linkedin.com/in/jenslaufer
KAGGLE	kaggle.com/jenslaufer
PUBLICATION	Towards Data Science — author

EDUCATION

11/1997
Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent)
University of Erlangen-Nuremberg
05/1989
Abitur (university entrance qualification)
Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018
Machine Learning Engineer Nanodegree
Udacity
Verify →
08/2017
Data Analyst Nanodegree
Udacity
Verify →